Cell Biology

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Lecture :3

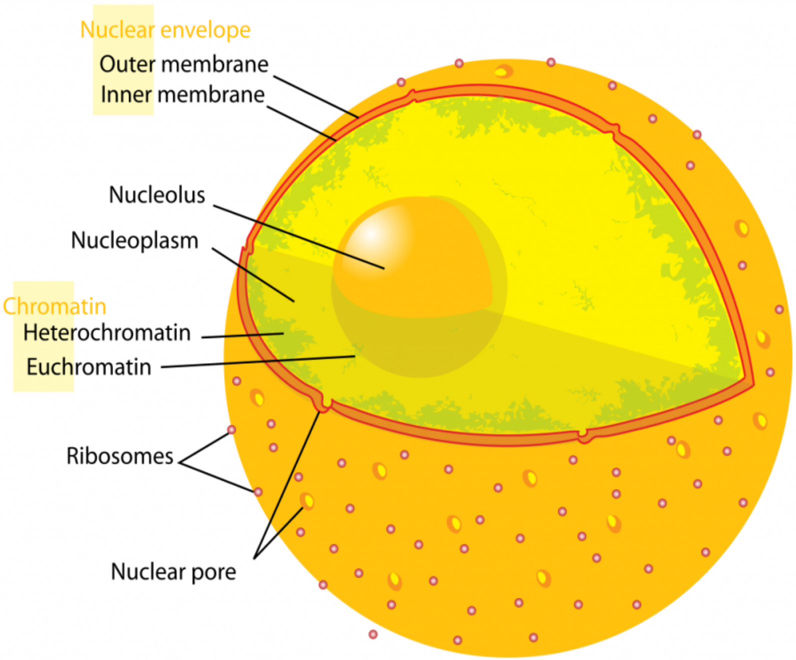
The Nucleus

Nucleus

- (Nuclei; from Latin nucleus or nuculeus, meaning kernel or seed)
- is a membrane-enclosed organelle found in eukaryotic cells.
- Eukaryotes usually have a single nucleus, but a few cell types, such as
- mammalian red blood cells, have no nuclei, and a few others have many.
- The nucleus is the largest cellular organelle in animal cells.
- In mammalian cells, the average diameter of the nucleus is
- approximately **6 micrometres (µm**), which occupies about 10% of the
- total cell volume. The viscous liquid within it is called nucleoplasm, and is similar in composition to the cytosol found outside the nucleus.
- It appears as a dense, roughly spherical or irregular organelle.
- The composition by dry weight of the nucleus is approximately: DNA 9%, RNA 1%, Histone Protein 11%, Residual Protein 14%, Acidic Proteins 65%.

- The nucleus controls protein synthesis in the cytoplasm by sending molecular messenger in the form of RNA, this messenger RNA (mRNA), is synthesized in the nucleus, then the mRNA conveys the genetic messages to the cytoplasm via the nuclear pores.
- Once in the cytoplasm the mRNA attaches to ribosomes, and the genetic message is translated into the primary structure of specific protein.
- The nucleus contains most of the genes that control the cell (some genes located in mitochondria and chloroplast). It is generally the most conspicuous organelles in a eukaryotic cell.

- Nucleus mainly composed of four different parts:
- **a-Nuclear envelope.**
- **b-Nucleoplasm.**
- **c-Chromatin.**
- **d-Nucleolus.**


Nuclear envelope
Outer membrane
Inner membrane
Nucleolus
Nucleoplasm
Chromatin
Heterochromatin
Euchromatin
Ribosomes
Nuclear pore

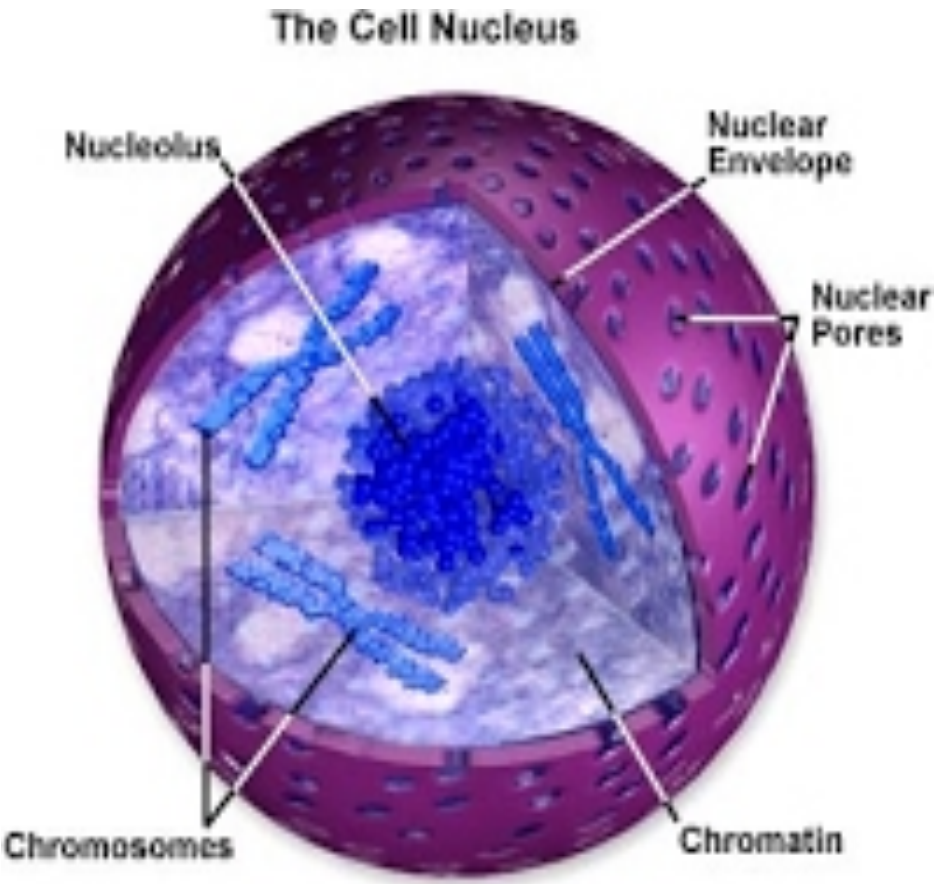
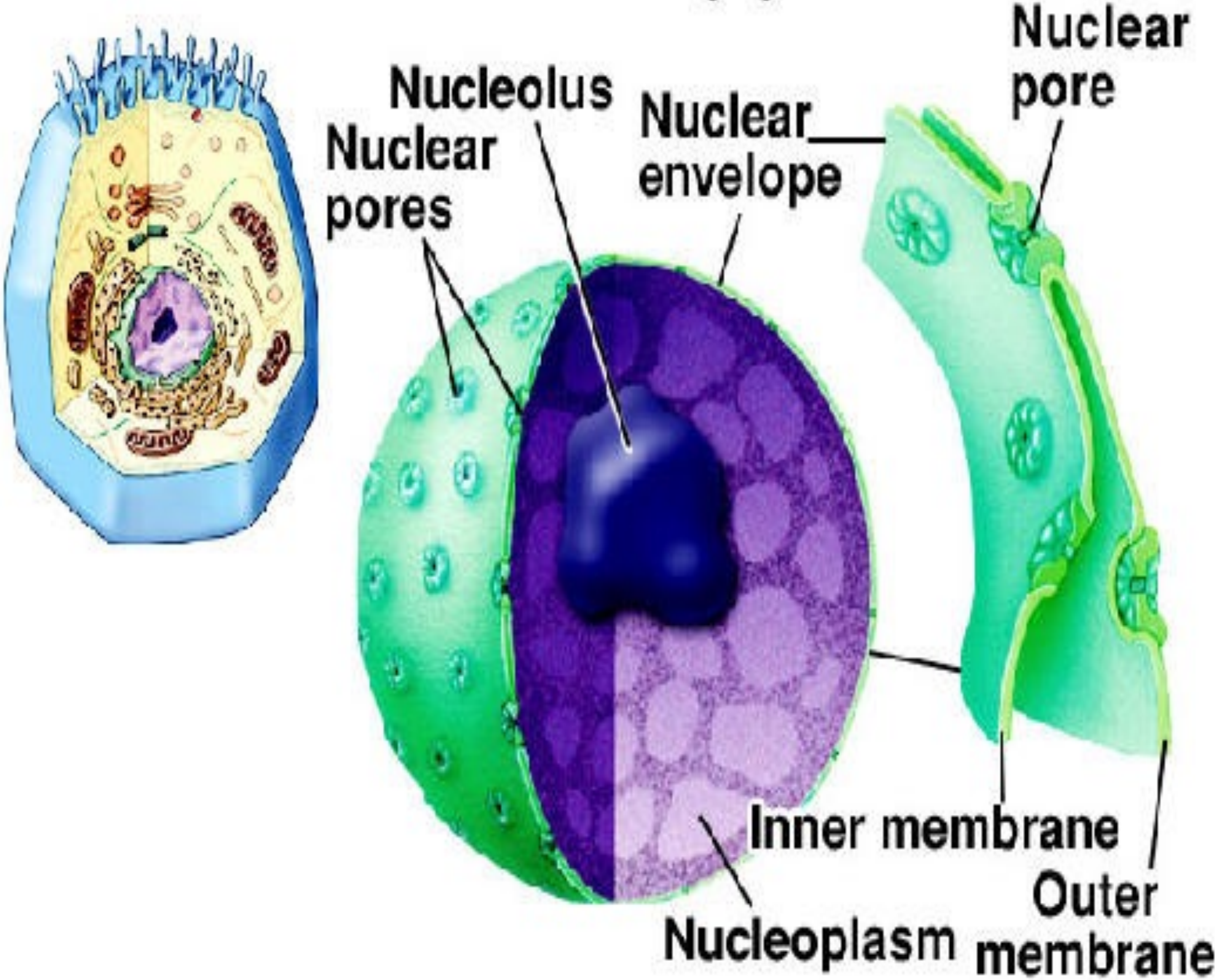

The Cell Nucleus
Nucleolus
Nuclear Envelope
Nuclear Pores
Chromosomes
Chromatin

Figure 1

Nucleus (1)


Nucleolus
Nuclear pores
Nuclear envelope
Nuclear pore
Inner membrane
Outer membrane
Nucleoplasm

a-Nuclear envelope

- The nuclear envelope, otherwise known as nuclear membrane,
- consists of two cellular membranes, an inner and an outer
- membrane, arranged parallel to one another and separated by 20 to 40 nanometres (nm).
- The nuclear envelope completely encloses the nucleus and separates the cell's genetic material from the surrounding cytoplasm, serving as a barrier to prevent macromolecules from diffusing freely between the nucleoplasm and the cytoplasm.
- The outer nuclear membrane is continuous with the membrane of the rough endoplasmic reticulum (RER), and is similarly studded with ribosomes.
- The space between the membranes is called the perinuclear space and is continuous with the RER lumen.

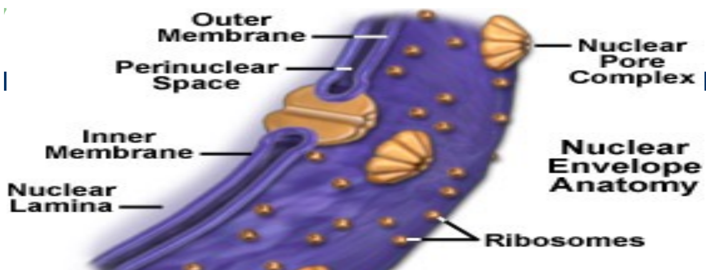
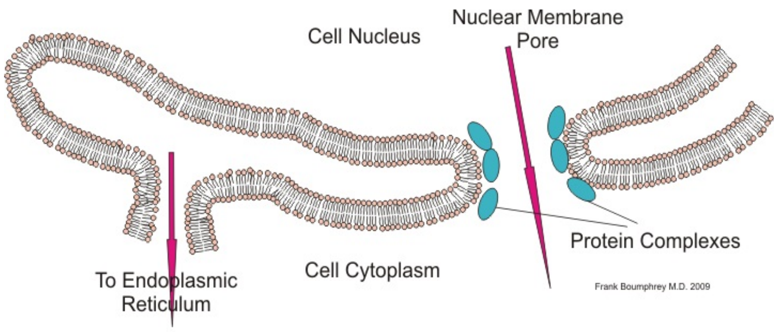
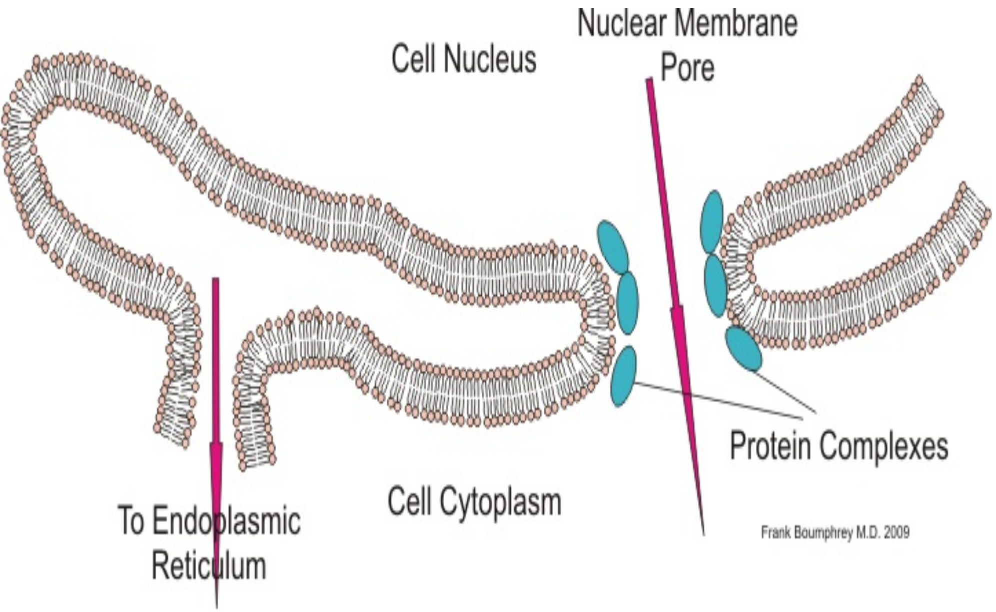
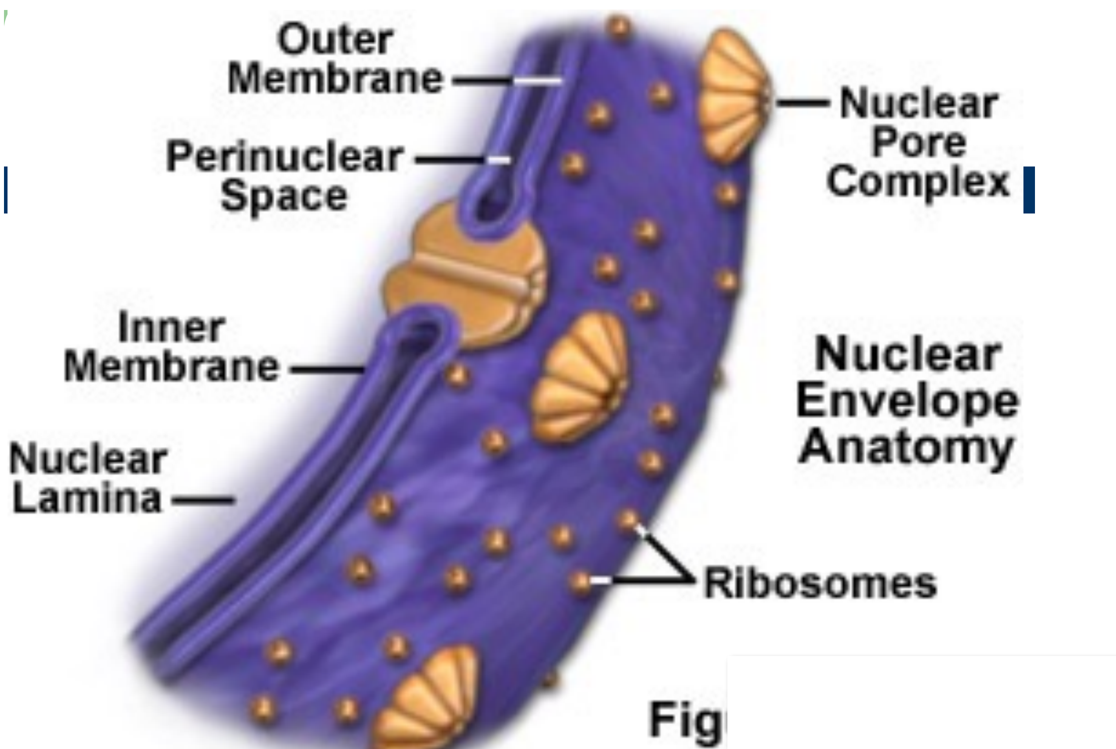


Figure 1

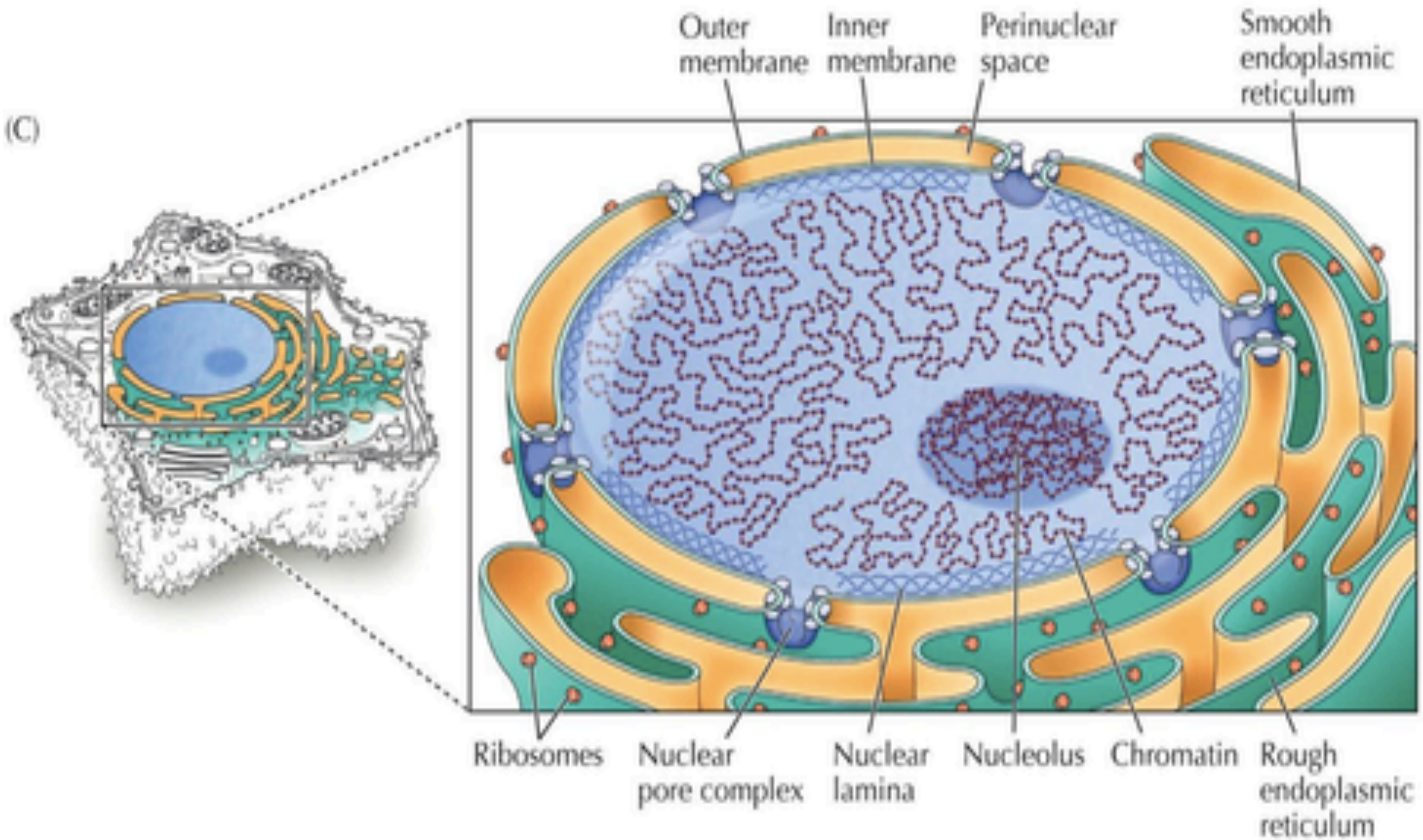
- -The nuclear envelope is punctured by thousands of nuclear pores, large protein complexes.
- There are average 1000 nuclear pore complexes in the nuclear envelope of a vertebrate cell, but it varies depending on cell type and stage in the cell cycle.
- -Nuclear pore complexes allow the transport of molecules across the
- nuclear envelope, this transport includes RNA and ribosomal proteins
- moving from nucleus to the cytoplasm and proteins (such as DNA
- polymerase), carbohydrate, lipids …… moving into the nucleus.
- -The pores are about 125 million daltons in molecular weight and consists of around 50 (in yeast) to several hundred proteins (in vertebrates).
- -The pores are 100nm in total diameter: however, the gap through which molecule freely diffuse is only about
- 9nm wide, due to the presence of regulatory
- system within the center of the pore.




Nuclear Membrane Pore
Cell Nucleus
Protein Complexes
To Endoplasmic Reticulum
Cell Cytoplasm
Frank Boumphrey M.D. 2009

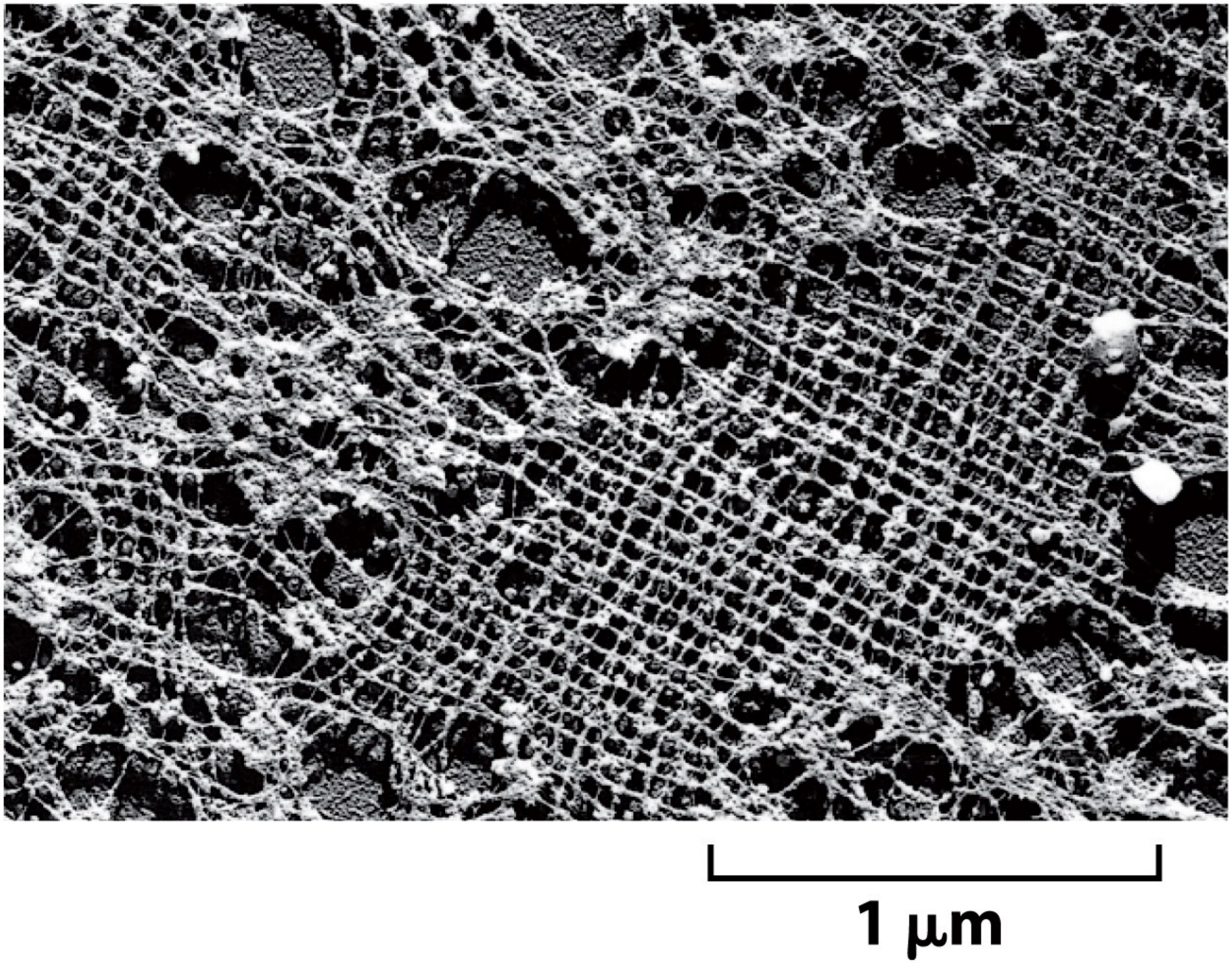

Outer Membrane
Perinuclear Space
Nuclear Pore Complex
Inner Membrane
Nuclear Envelope Anatomy
Nuclear Lamina
Ribosomes
Fig

Nuclear lamina (the major component of nucleoskeleton) is attached to the inside surface of the inner nuclear membrane

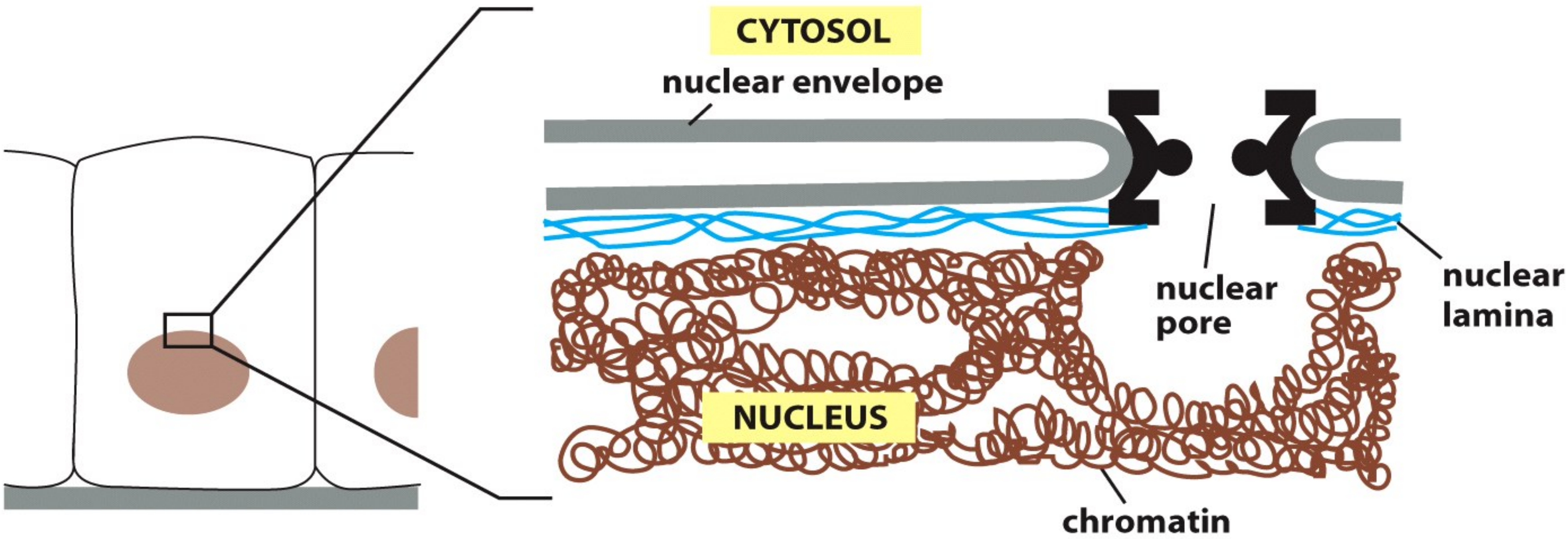


THE CELL: A MOLECULAR APPROACH 6e, Figure 9.1 (Part 3)

Electron microscopic image of the nuclear lamina



The function of nuclear lamina: to stabilize nuclear envelope and to bind chromatin



The nuclear lamina consists of 3 types of lamin proteins: A, B, and C.

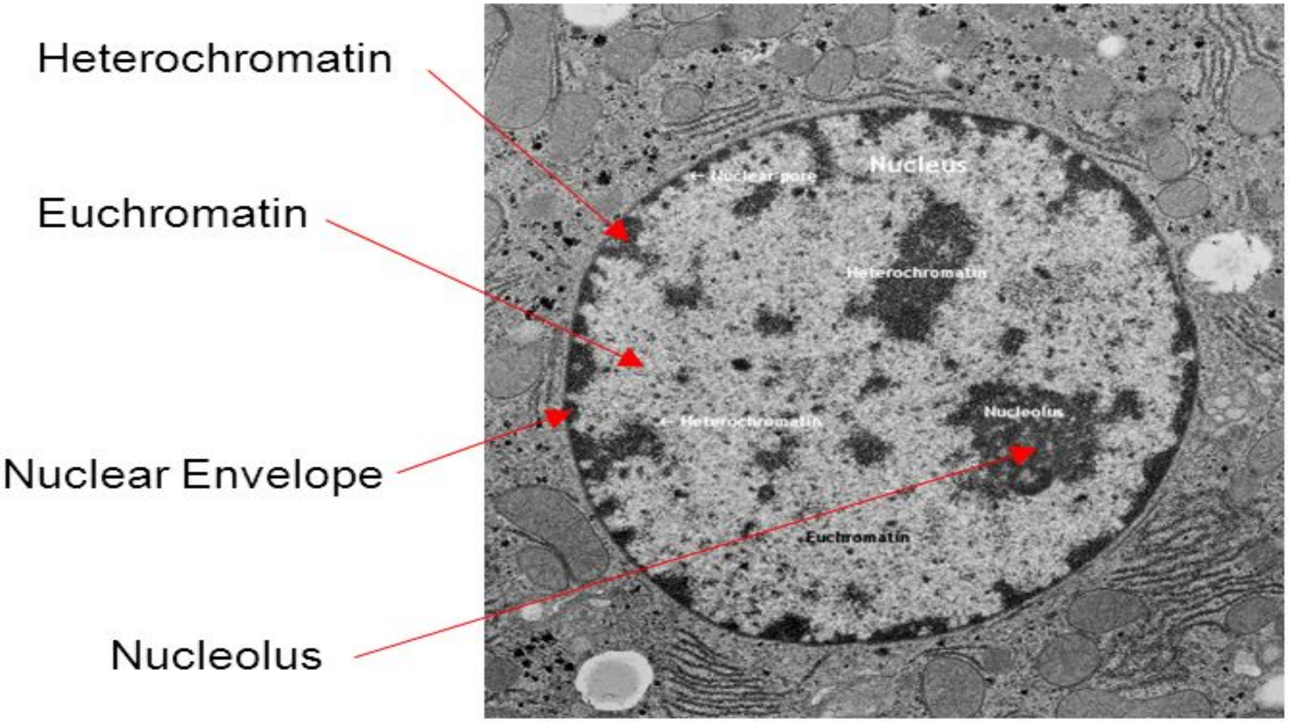
Lamin A and C bind chromatin, whereas lamin B binds them to the nuclear membrane via lamin B receptor, an integral membrane protein.

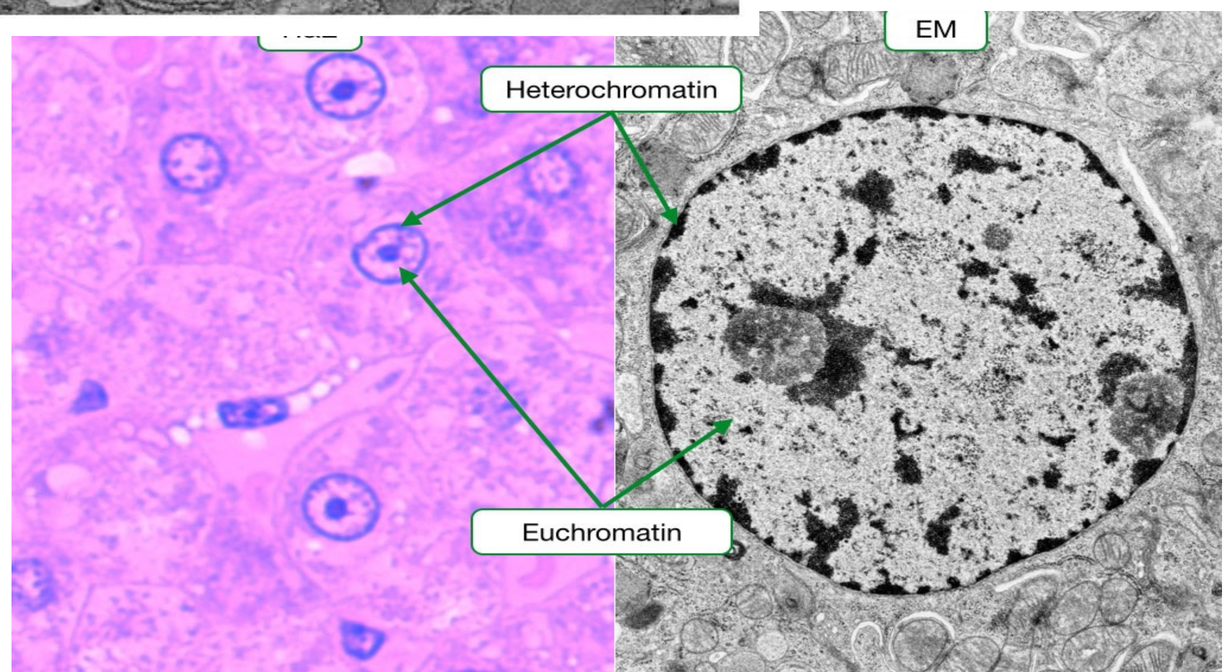
Phosphorylation of lamins leads to the disassembly of the structure during cell division.

A **mutation of lamin A causes Hutchinson-Gilford progeria**, an early aging syndrome.

b-Chromatin

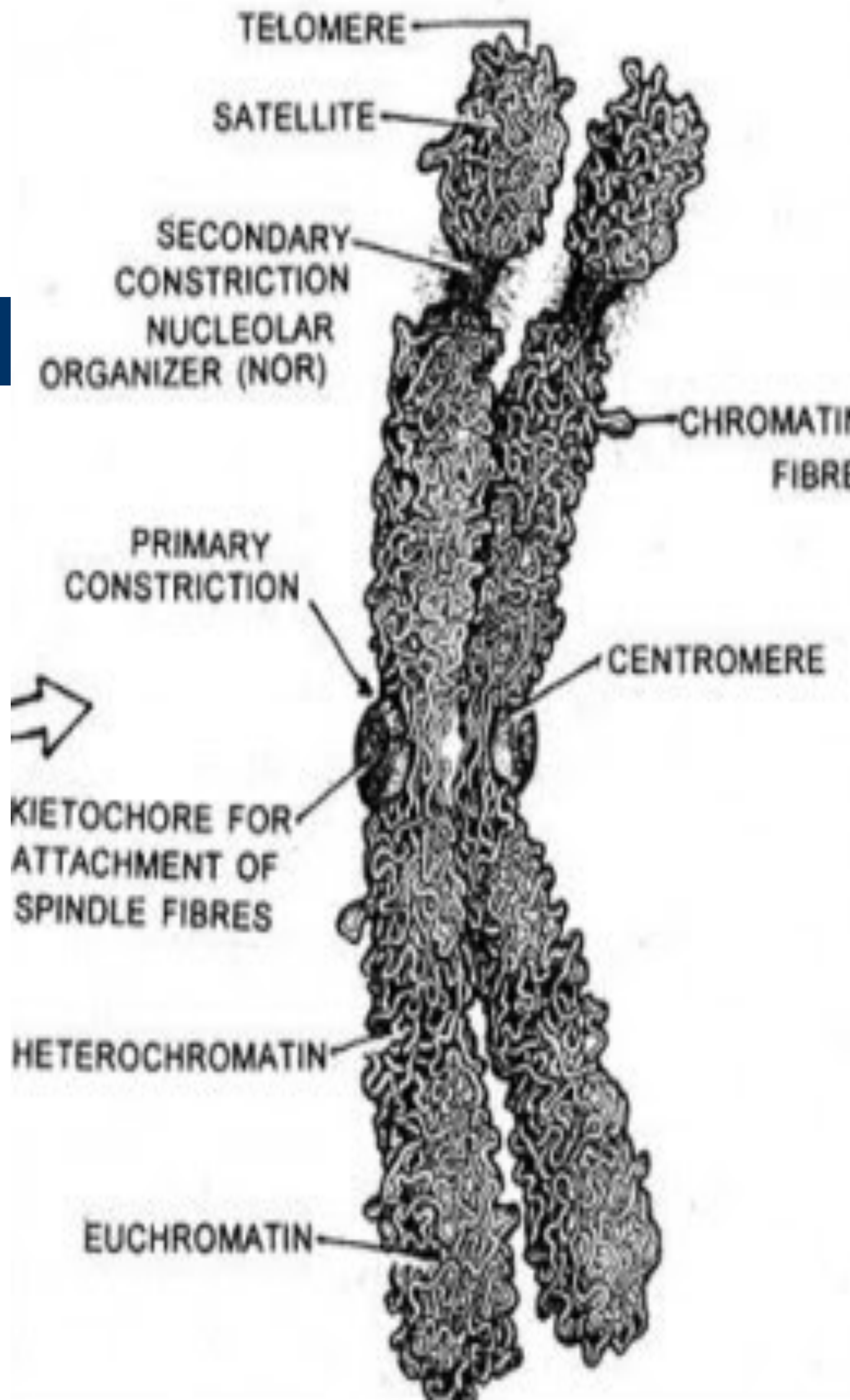
- Cell nuclei contain most of the cell's genetic material,
- organized as multiple long linear DNA molecules in
- complex with a large variety of proteins, such as
- histones, to form chromosomes.
- The genes within these chromosomes are the cell's nuclear genome and are structured in such a way to promote cell
- function.
- The nucleus maintains the integrity of genes and controls the activities of the cell.


Heterochromatin
Euchromatin
Nuclear Envelope
Nucleolus
Nucleus
← Nuclear pore
Heterochromatin
← Heterochromatin
Nucleolus
Euchromatin

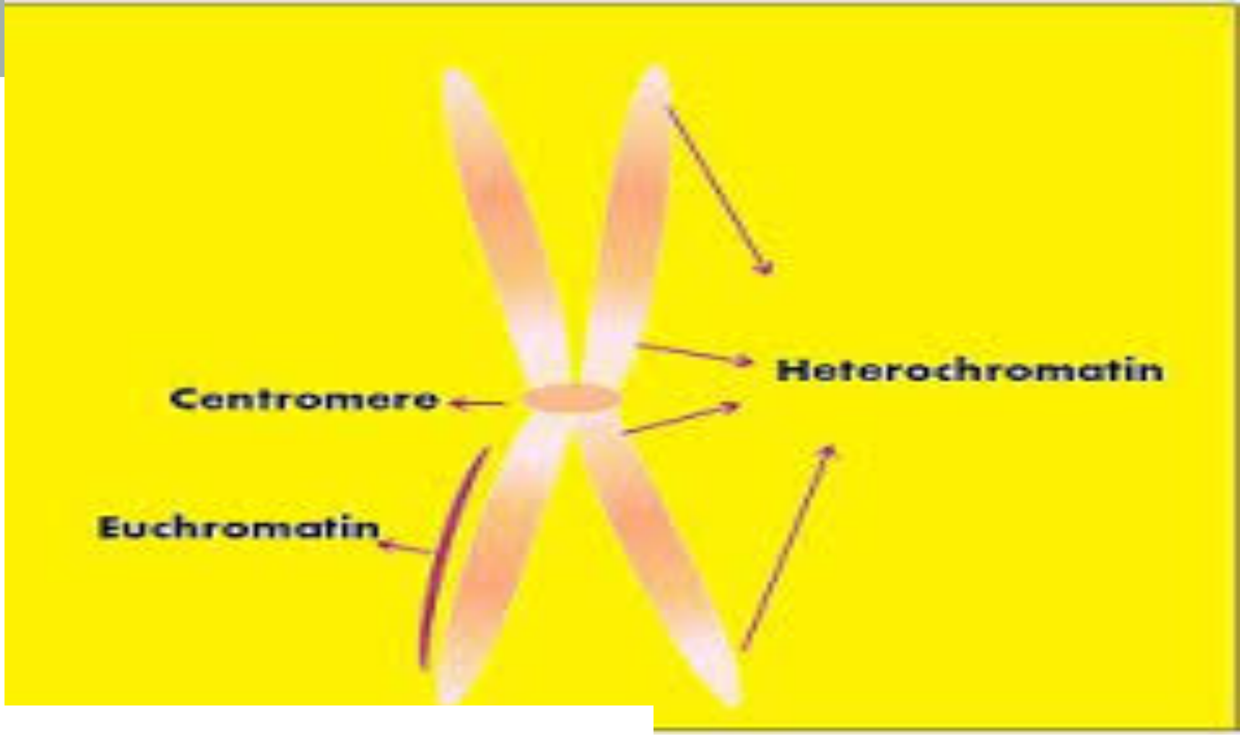

EM
Heterochromatin
Euchromatin

- Chromatin is a complex of macromolecules found in nucleus,
- consisting of DNA, protein, and RNA.
- The primary functions of chromatin are
- **1) to package DNA into a more compact, denser shape,**
- **2) to reinforce the DNA macromolecule to allow mitosis,**
- **3) to prevent DNA damage, and**
- **4) to control gene expression and DNA replication**.
- The primary protein components of chromatin are histones that
- compact the DNA. Chromatin is only found in eukaryotic cells.
- The structure of chromatin depends on the stage of the cell
- cycle.

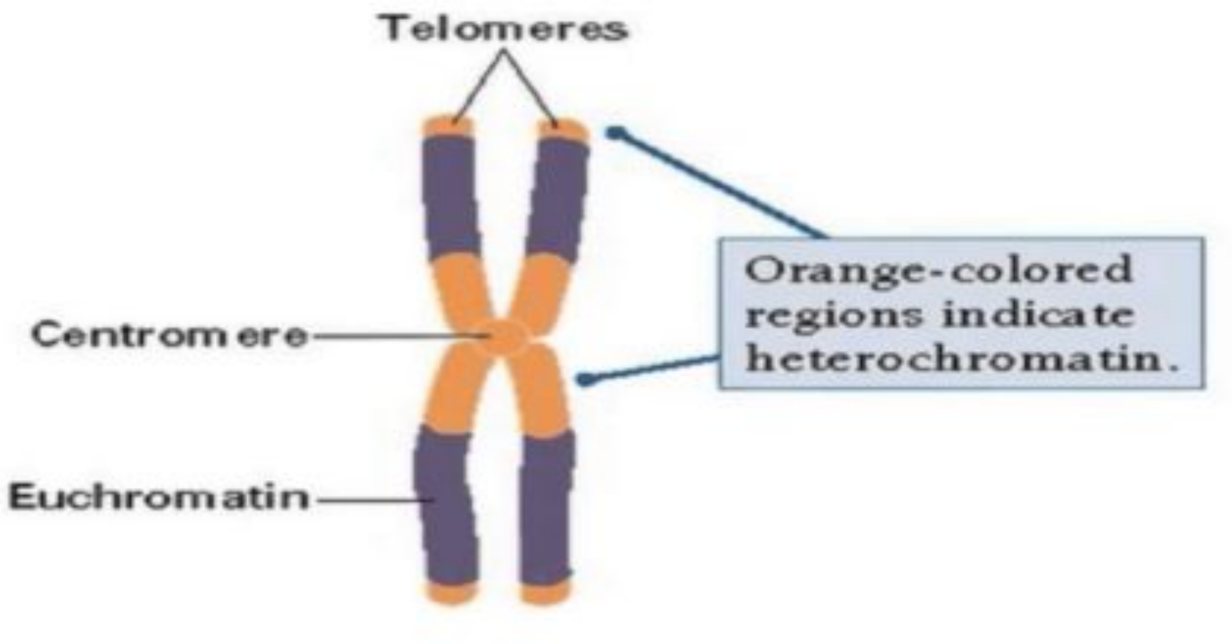
- During interphase, the chromatin is structurally loose
- to allow access to RNA and DNA polymerases that
- transcribe and replicate the DNA.
- -The DNA which codes genes that are actively
- transcribed ("turned on") is more loosely packaged
- and associated with RNA polymerases (referred to as
- euchromatin) while that DNA which codes inactive
- genes ("turned off") is more condensed and
- associated with structural proteins (heterochromatin).


TELOMERE
SATELLITE
SECONDARY CONSTRICTION
NUCLEOLAR ORGANIZER (NOR)
CHROMATIN FIBRE
PRIMARY CONSTRICTION
CENTROMERE
KIETOCHORE FOR ATTACHMENT OF SPINDLE FIBRES
HETEROCHROMATIN
EUCHROMATIN

- Euchromatin
- *Less condensed region of chromosomes.
- **Transcriptionally active.
- Heterochromatin
- *Tightly compacted region of chromosomes
- **Transcriptionally inactive.

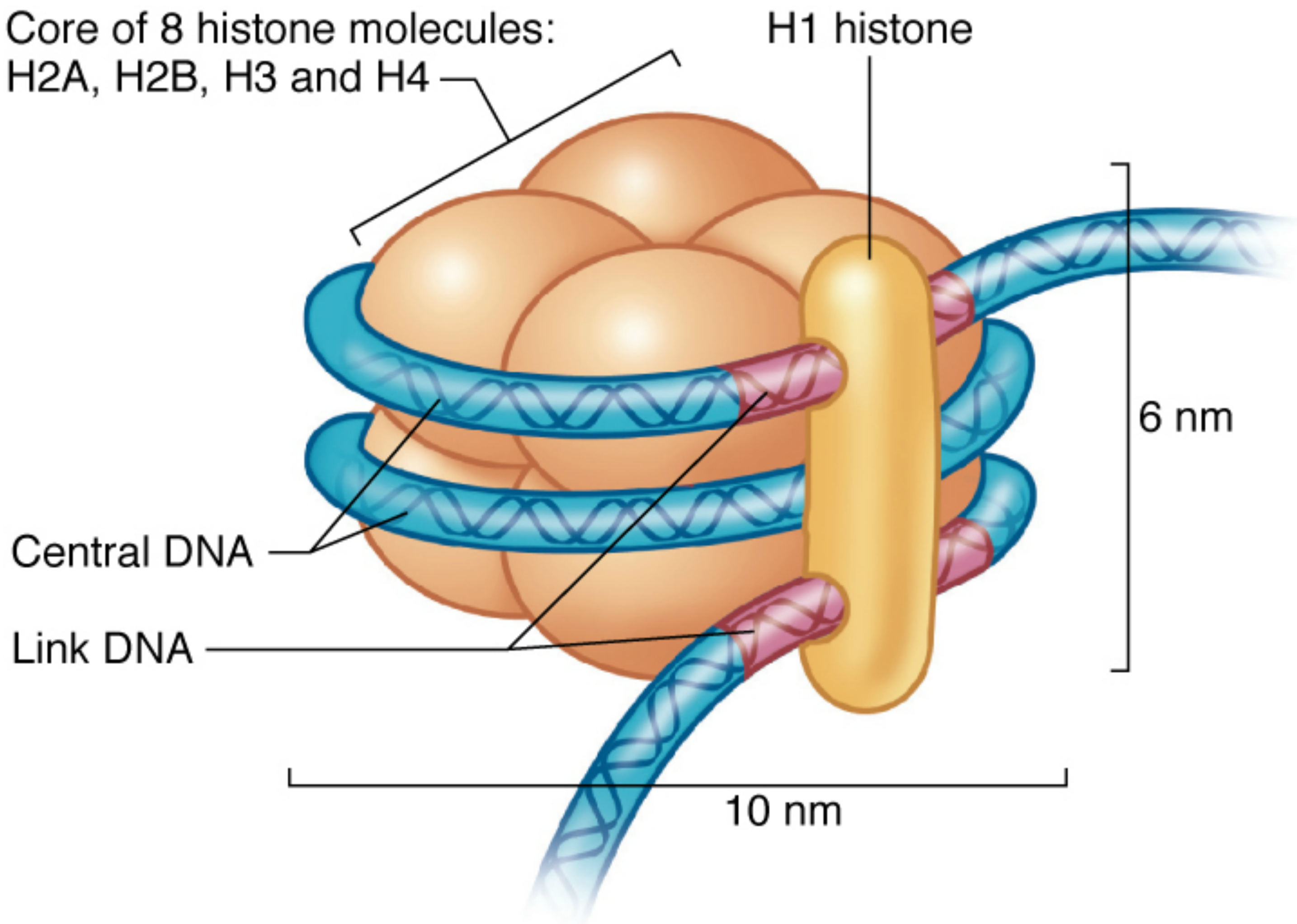


Heterochromatin
Centromere
Euchromatin

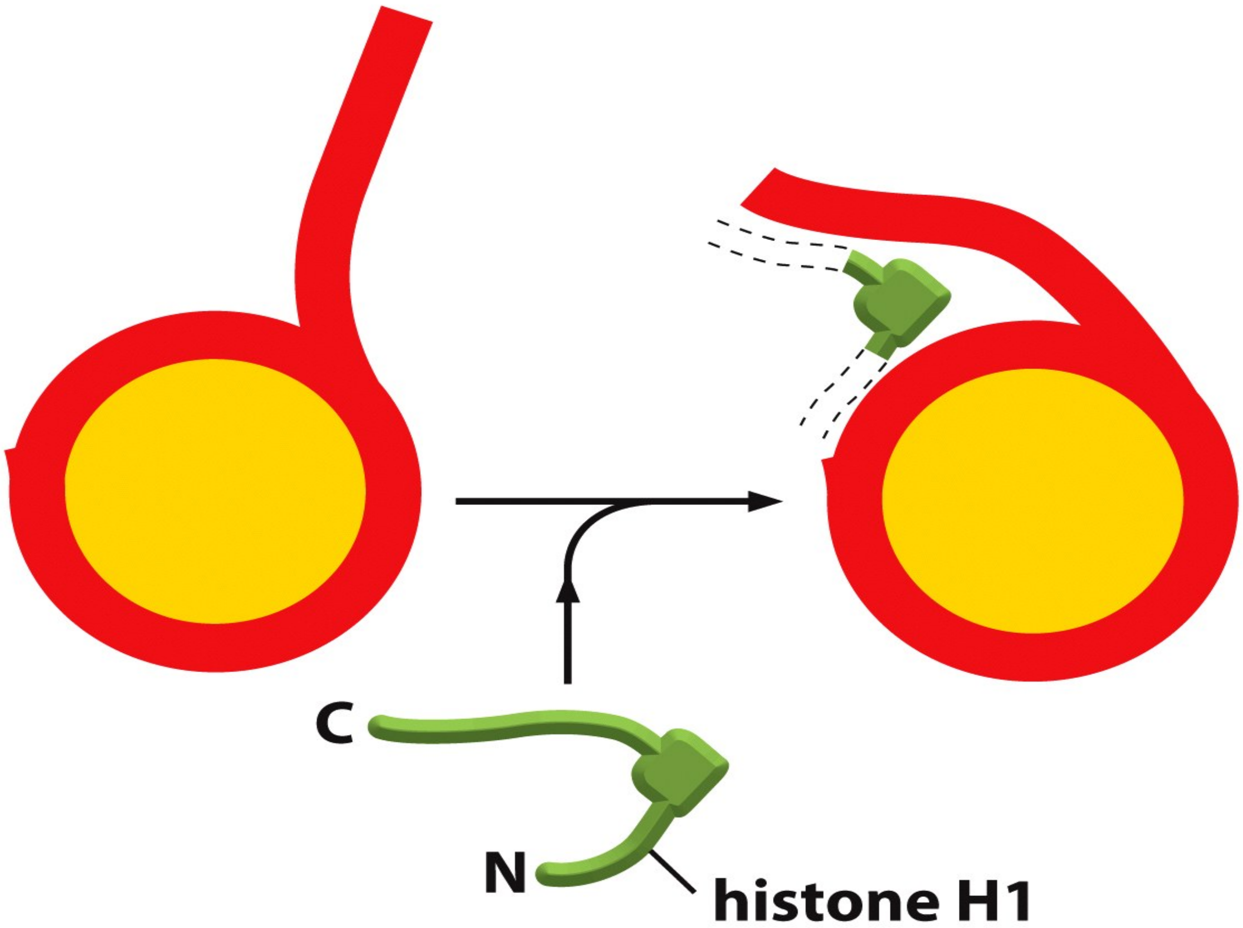


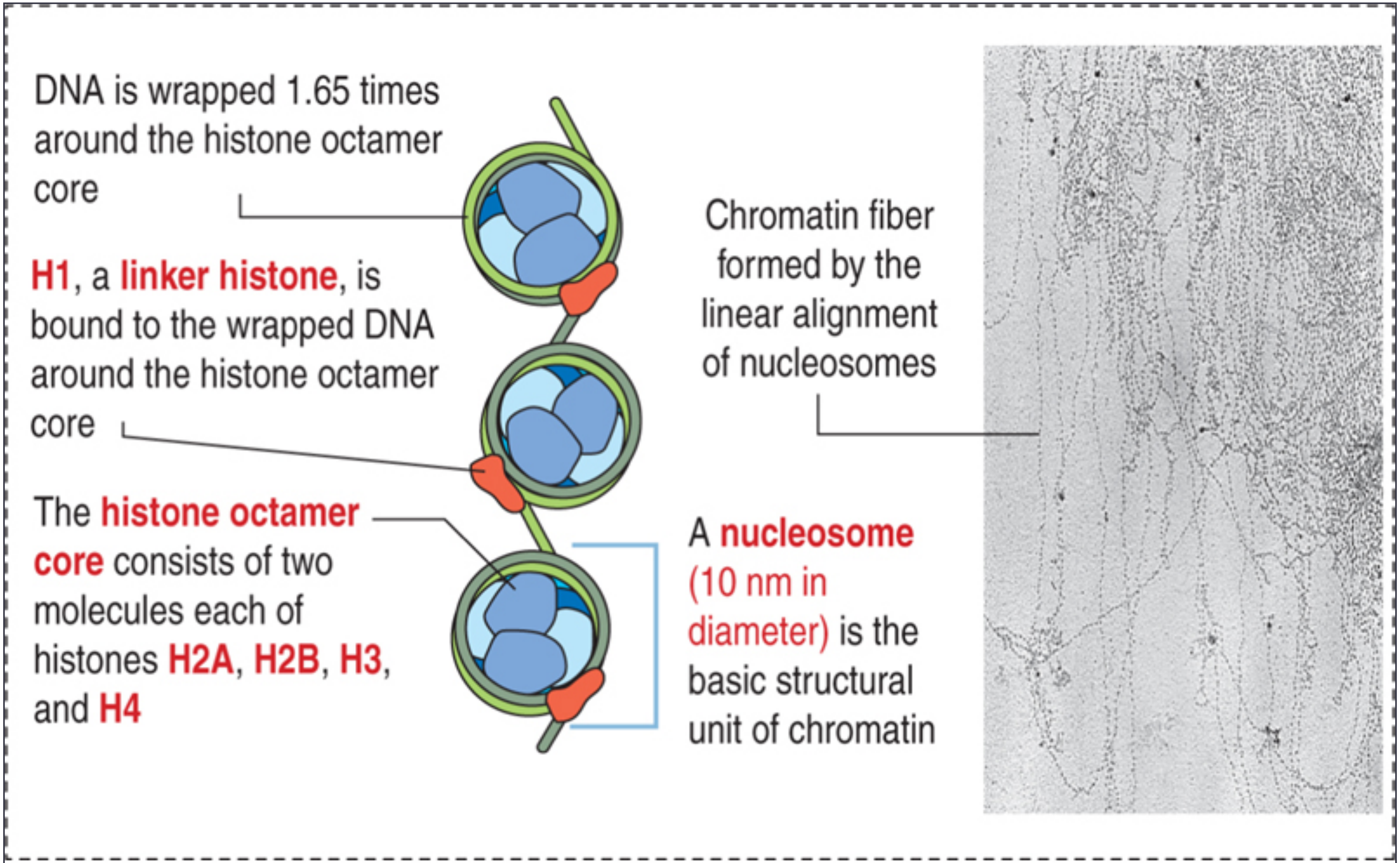
Telomeres
Centromere
Euchromatin
Orange-colored regions indicate heterochromatin.

The basic organizational unit of DNA in chromosomes: the nucleosome



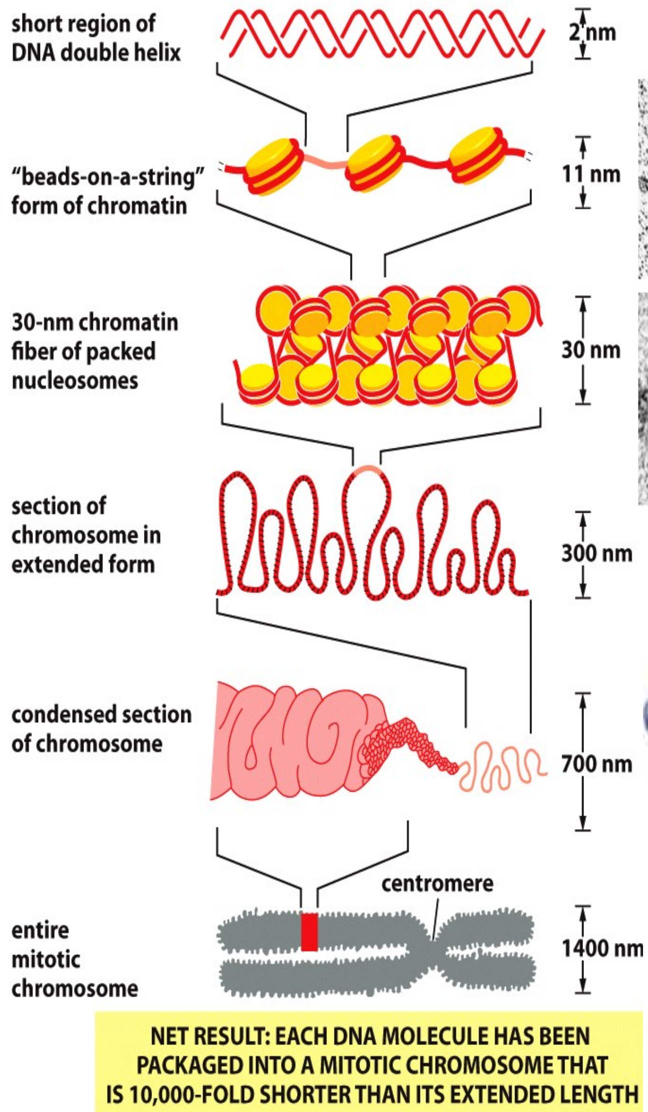
A linker histon (H1) determines the angle how linker DNA leaves the core of the nucleosomes allowing further condensation

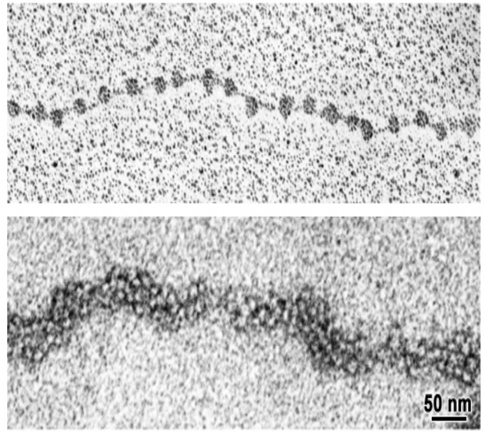


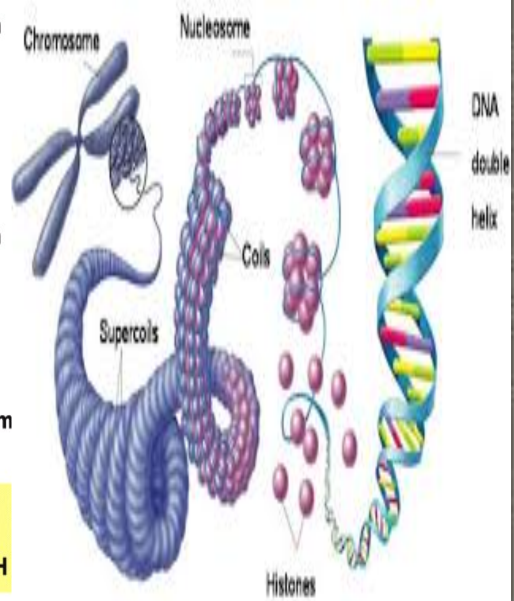


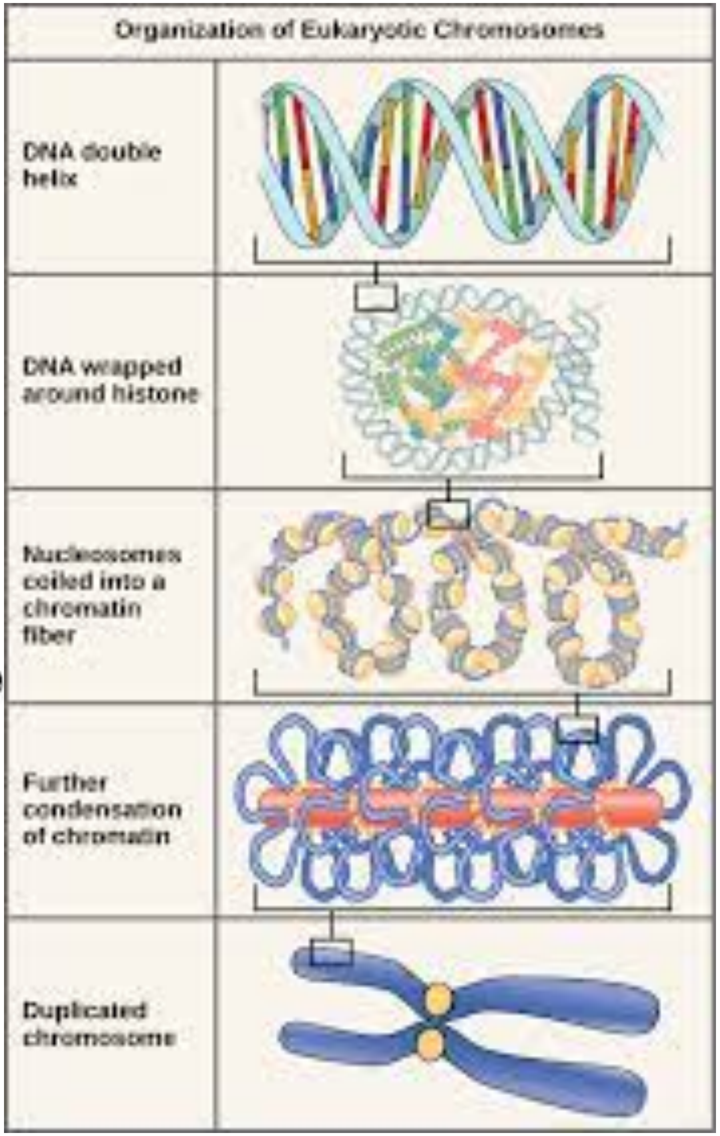
Kierszenbaum & Tres: Histology and Cell Biology: An Introduction to Pathology, 3e

Different levels of DNA packing

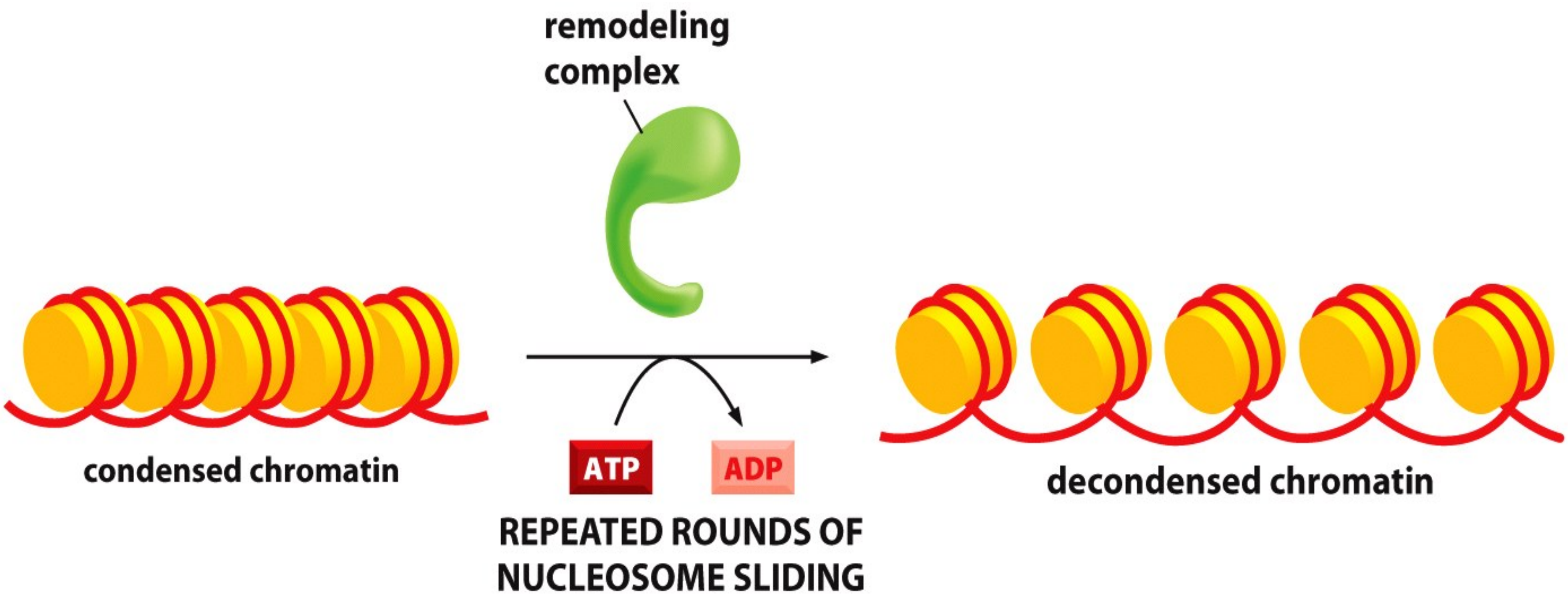






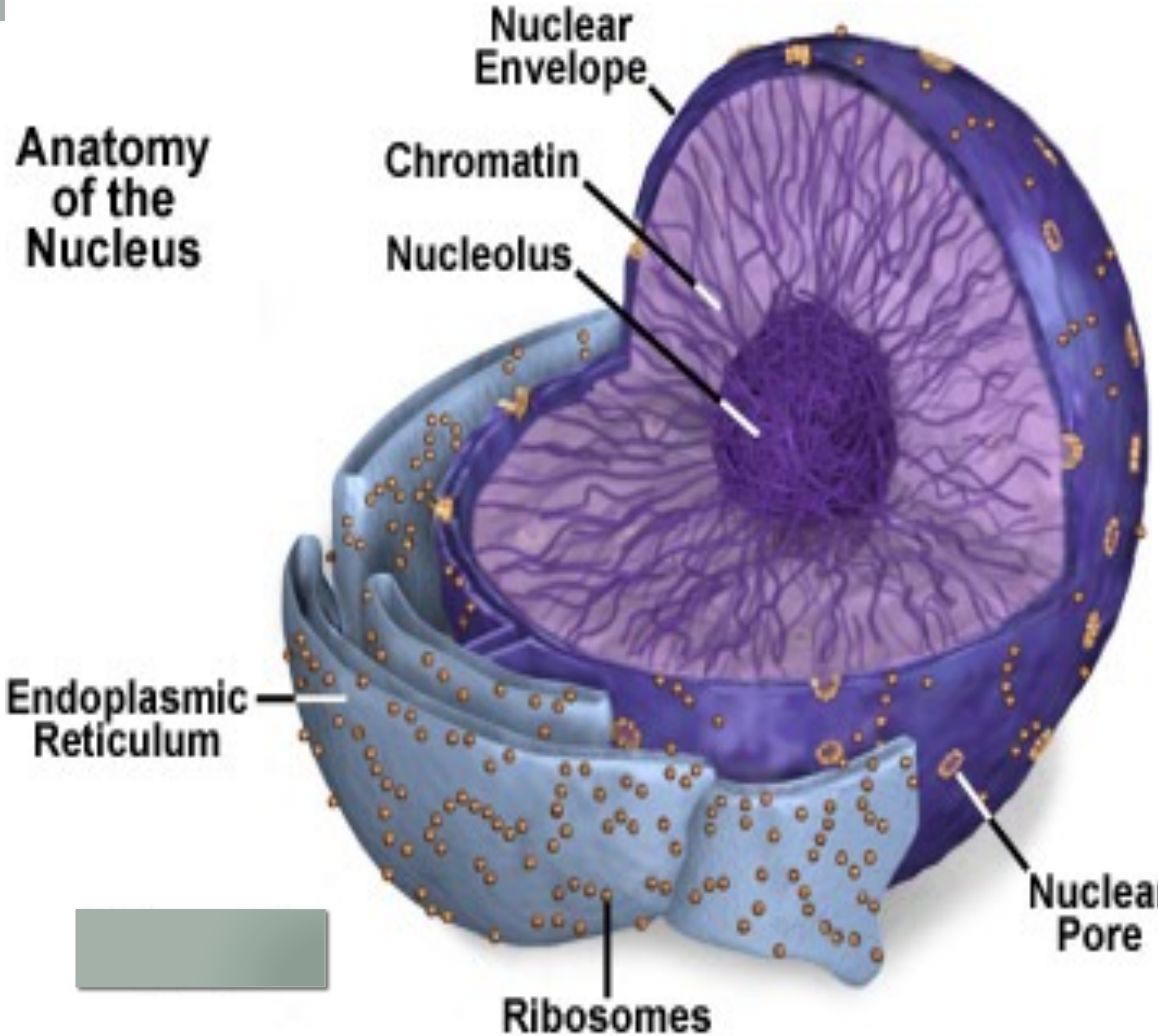


Chromatin-remodeling complexes decondense the chromatin making it accessible to other proteins for replication, transcription and their regulations



c-nucleolus

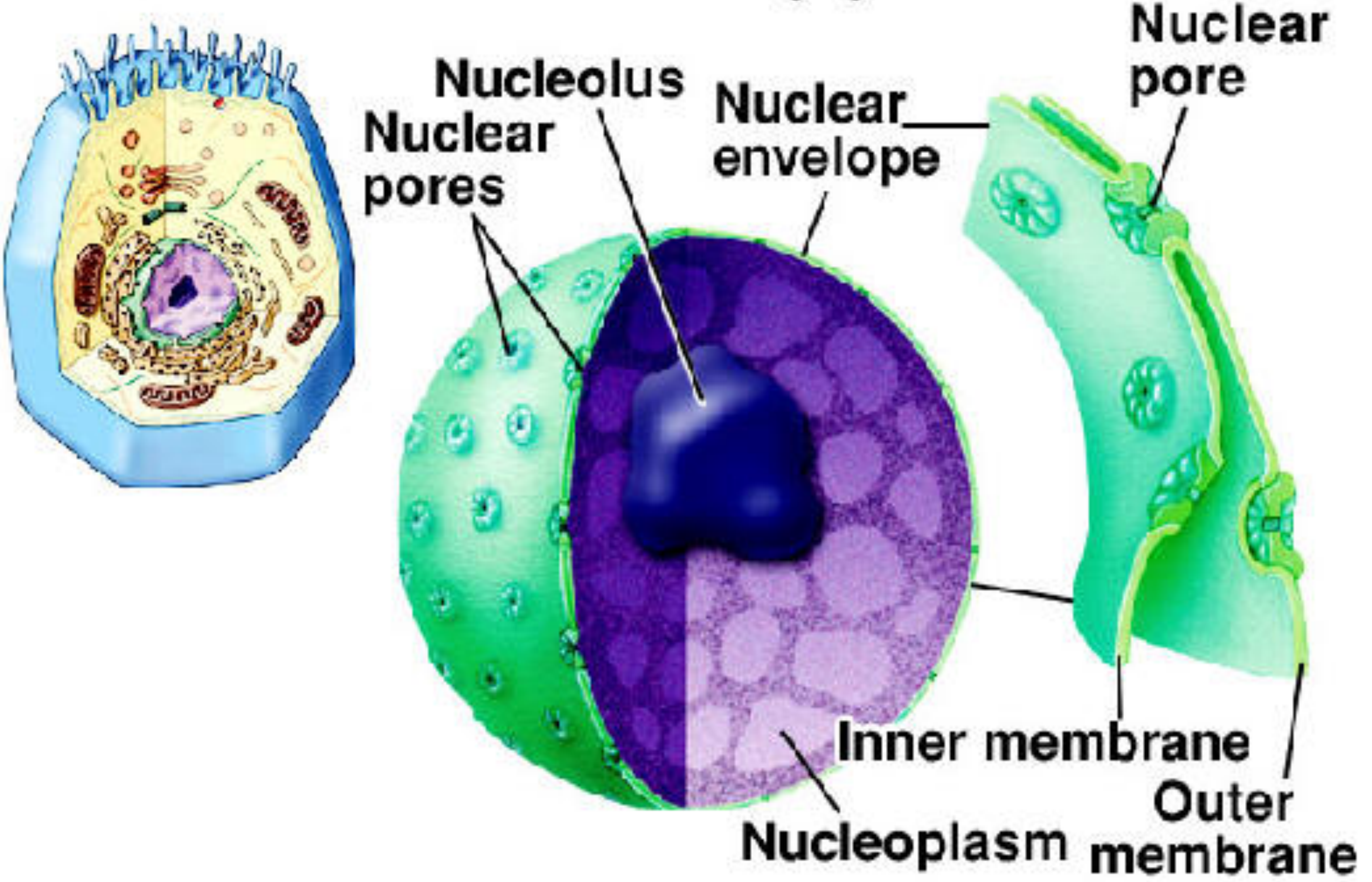
- The most visible structure within the non dividing nucleus
- is the nucleolus, Some times there are two or more
- nucleoli, the number depends on the species and the
- stage in the cells reproductive cycle.
- The nucleolus is roughly spherical the most prominent structure within the nucleus is the nucleolus, which is the site of rRNA transcription, The nucleolus, is not surrounded by a membrane and its rich in (rRNA and protein).


Anatomy of the Nucleus
Nuclear Envelope
Chromatin
Nucleolus
Endoplasmic Reticulum
Ribosomes
Nuclear Pore

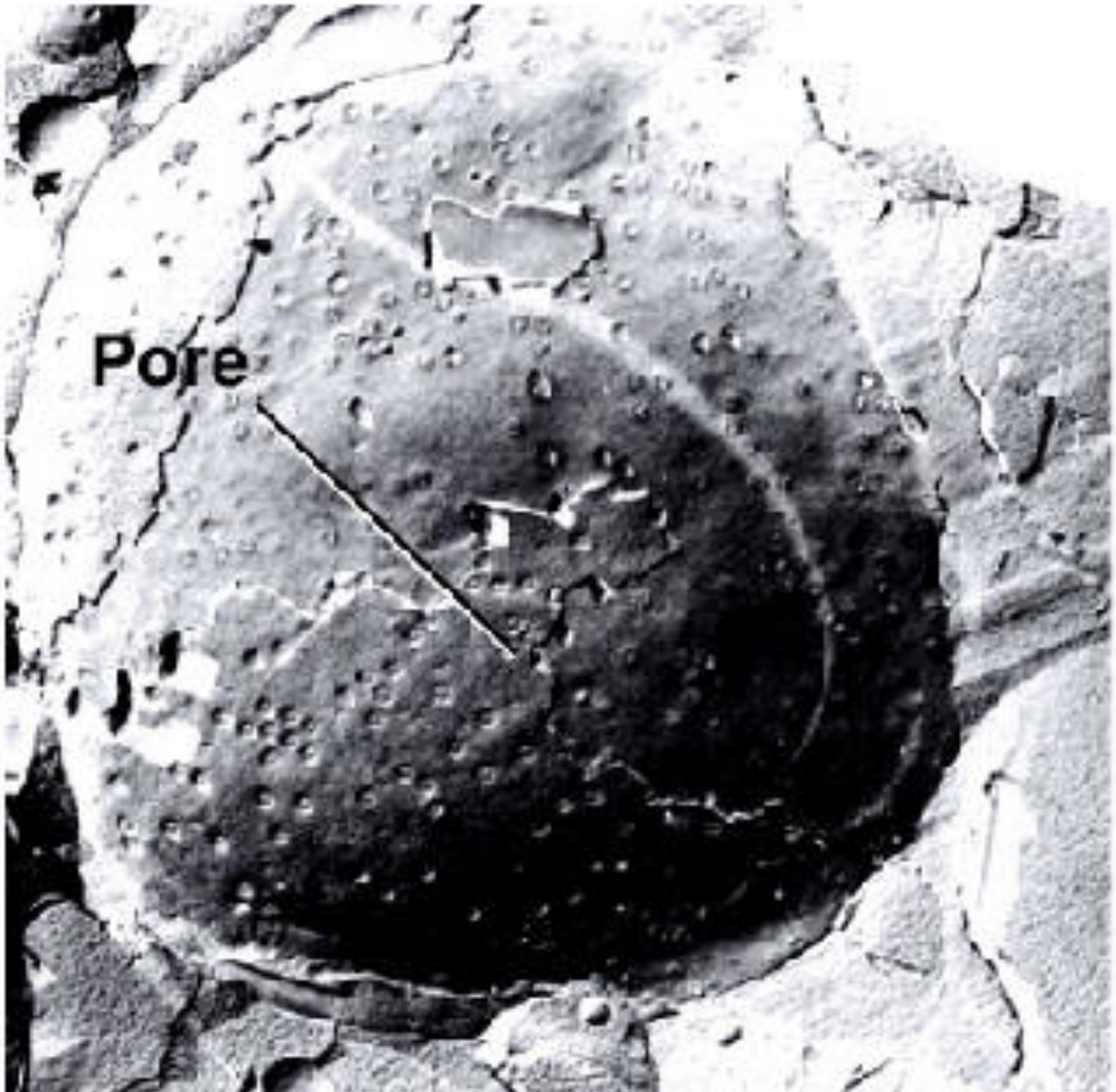
d-Nucleoplasm

- Is the internal cytoplasm within the nucleus, that chromatin and
- nucleolus are situated inside it.
- Similar to the cytoplasm of a cell, the nucleus contains nucleoplasm,
- karyoplasm, or nucleus sap. The nucleoplasm is one of the types of
- protoplasm, and it is enveloped by the nuclear membrane or nuclear
- envelope.
- The nucleoplasm includes the chromosomes and nucleoli. Many
- substances such as nucleotides (necessary for purposes such as the
- replication of DNA) and enzymes (which direct activities that take
- place in the nucleus) are dissolved in the nucleoplasm. The soluble,
- liquid portion of the nucleoplasm is called the nucleosol or nuclear
- hyaloplasm.
- *The nucleoplasm is the site of various nuclear events and activities. For instance, it is where gene expression, DNA replication, and DNA repair occurs.

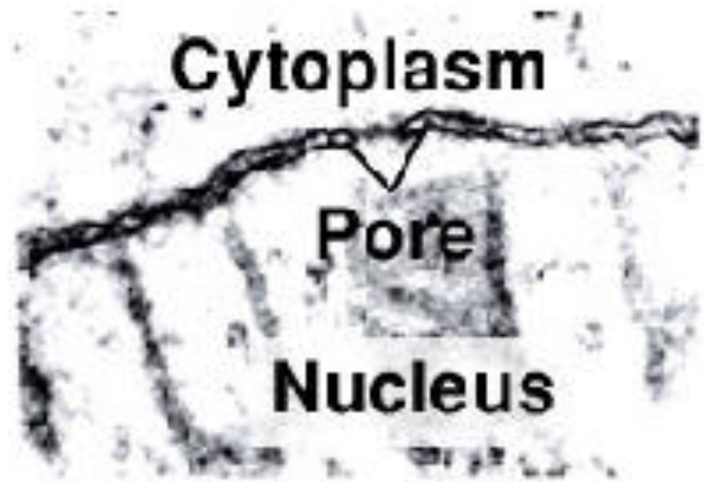
Nucleus (1)


Nucleolus
Nuclear pores
Nuclear envelope
Nuclear pore
Inner membrane
Outer membrane
Nucleoplasm

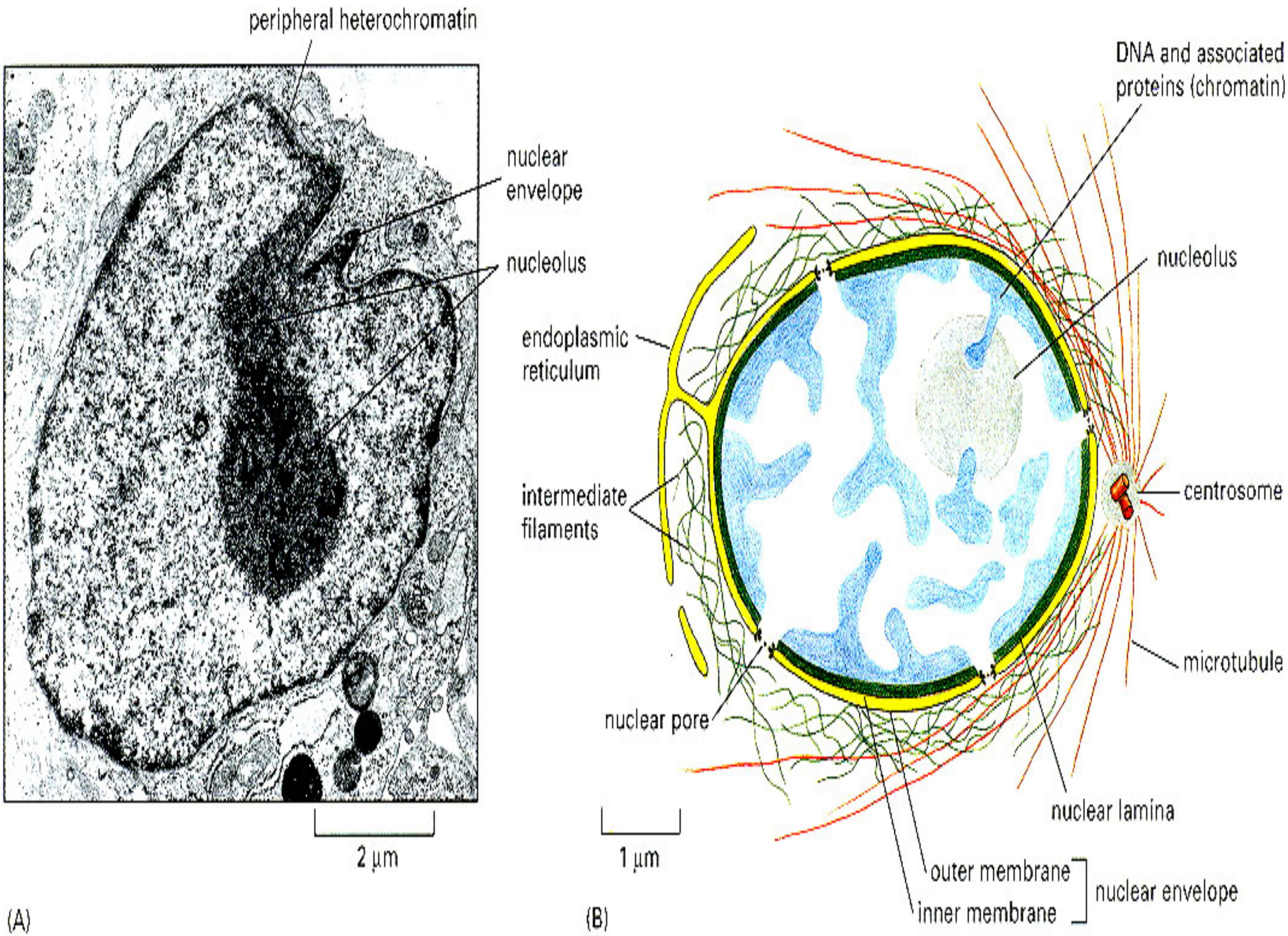
Nucleus (2)



Cell nucleus



Nuclear membrane


peripheral heterochromatin
nuclear envelope
nucleolus
2 µm
(A)
DNA and associated proteins (chromatin)
nucleolus
endoplasmic reticulum
intermediate filaments
centrosome
microtubule
nuclear pore
nuclear lamina
1 µm
outer membrane
inner membrane
nuclear envelope
(B)


Thank
You